

ST PETE-CLEARWATER, FL, USA

WMO: 722116

Lat: 27.911N

Lon: 82.688W

Elev: 3

StdP: 101.28

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12873

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 5.6	(c) 7.5	(d) -3.3	(e) 2.9	(f) 9.2	(g) -1.0	(h) 3.5	(i) 11.0	(j) 11.3	(k) 18.8	(l) 10.3	(m) 17.7	(n) 4.1	(o) 20	(p) 0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 7.1	(c) 33.6	(d) 25.2	(e) 32.9	(f) 25.2	(g) 32.4	(h) 25.2	(i) 27.2	(j) 30.3	(k) 26.7	(l) 30.2	(m) 26.3	(n) 30.1	(o) 3.7	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	28.6	26.0	21.3	28.5	25.2	20.4	28.2	85.6	29.9	83.1	30.1	81.5	30.0	31.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.4	8.4	7.5	1.5	34.8	2.7	0.8	-0.5	35.4	-2.1	35.9	-3.6	36.3	-5.6	36.9
			-0.8	28.0	2.3	1.0	-2.5	28.8	-3.8	29.4	-5.1	29.9	-6.8	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.4	16.3	18.1	20.1	23.0	25.9	27.8	28.3	28.6	27.9	25.1	20.9	18.3	(6)
	DBStd	5.18	4.38	4.06	3.50	2.56	1.78	1.31	1.24	1.21	1.41	2.75	3.34	4.26	(7)
	HDD10.0	9	6	2	0	0	0	0	0	0	0	0	0	2	(8)
	HDD18.3	241	92	49	24	3	0	0	0	0	0	2	16	55	(9)
	CDD10.0	4894	201	229	314	388	494	535	567	575	536	468	326	261	(10)
	CDD18.3	2083	28	44	80	141	236	285	309	317	286	211	92	55	(11)
	CDH23.3	18745	102	183	396	900	2060	3003	3408	3551	2895	1639	408	201	(12)
	CDH26.7	6432	6	19	55	206	697	1163	1334	1403	1024	450	58	17	(13)
Wind	WSAvg	3.7	4.0	4.0	4.3	4.2	3.8	3.4	3.0	3.1	3.5	3.9	3.8	3.8	(14)
Precipitation	PrecAvg	1276	64	68	77	61	64	176	213	219	170	65	39	60	(15)
	PrecMax	1769	139	295	232	219	226	357	385	346	346	167	123	396	(16)
	PrecMin	981	10	11	8	5	6	78	101	118	44	9	6	13	(17)
	PrecStd	203	39	59	60	43	54	76	66	70	80	35	29	96	(18)
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.4	28.6	29.2	31.2	33.0	34.3	34.1	34.2	33.9	32.6	29.7	28.0	(19)
		MCWB	21.4	21.4	21.5	22.0	23.3	25.0	25.3	25.8	25.1	24.8	23.3	22.2	(20)
	2%	DB	26.0	27.2	28.0	29.9	32.1	33.0	33.1	33.3	32.8	31.4	28.0	27.1	(21)
		MCWB	20.5	20.9	21.0	21.7	23.1	24.9	25.6	25.7	25.0	24.4	22.3	21.7	(22)
	5%	DB	24.3	25.9	27.2	28.9	31.2	32.4	32.5	32.7	32.2	30.3	27.2	25.9	(23)
		MCWB	19.5	20.1	20.5	21.5	22.9	24.9	25.6	25.7	25.1	24.0	21.6	21.2	(24)
	10%	DB	22.6	24.0	25.9	27.8	30.2	31.6	31.9	32.1	31.3	29.1	26.0	24.1	(25)
		MCWB	19.0	19.2	19.8	21.0	22.7	24.9	25.6	25.7	25.0	23.4	21.0	20.4	(26)
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.9	23.8	24.5	25.7	27.4	28.0	27.7	27.7	26.7	24.6	23.9	(27)
		MCDB	25.7	26.4	27.3	28.1	29.4	30.8	31.0	31.2	30.0	29.3	27.5	26.3	(28)
	2%	WB	21.6	21.9	22.9	23.7	25.0	26.5	27.1	27.1	26.8	25.7	23.9	22.9	(29)
		MCDB	24.1	25.3	26.0	27.5	28.8	30.5	30.3	30.6	29.7	28.9	26.6	25.4	(30)
	5%	WB	20.7	21.2	22.0	23.0	24.5	25.9	26.6	26.7	26.2	25.2	23.0	22.2	(31)
		MCDB	23.2	24.3	24.9	26.7	28.4	30.1	30.1	30.5	29.5	28.6	25.4	24.4	(32)
	10%	WB	19.6	20.4	21.1	22.3	24.0	25.6	26.0	26.2	25.7	24.6	22.1	21.2	(33)
		MCDB	22.2	23.2	24.1	26.0	28.0	29.8	29.8	30.2	29.2	27.9	24.7	23.5	(34)
Mean Daily Temperature Range		MDBR	9.3	9.3	9.1	9.0	8.7	7.6	7.3	7.1	7.0	7.5	8.3	8.5	(35)
	5% DB	MCDBR	9.9	9.9	9.7	9.5	9.4	8.4	7.8	7.8	7.8	7.6	8.4	8.7	(36)
		MCWBR	5.4	4.8	4.2	3.7	3.4	3.1	3.0	2.9	2.7	2.9	4.1	4.3	(37)
	5% WB	MCDBR	8.8	8.8	8.3	8.4	8.1	7.5	7.3	7.2	7.0	6.8	7.3	7.9	(38)
		MCWBR	5.7	5.1	4.9	4.4	3.5	3.1	3.3	3.1	2.9	3.2	4.3	4.9	(39)
Clear-Sky Solar Irradiance	taub	0.344	0.357	0.369	0.394	0.424	0.449	0.473	0.455	0.423	0.390	0.355	0.344	(40)	
	taud	2.525	2.496	2.449	2.343	2.314	2.324	2.264	2.329	2.384	2.462	2.506	2.550	(41)	
	Ebn at Noon	901	916	922	903	871	844	822	838	857	869	881	885	(42)	
	Edh at Noon	90	100	111	127	131	129	137	128	117	103	91	85	(43)	
All-Sky Solar Radiation	RadAvg	3.53	4.27	5.42	6.51	6.91	6.26	5.92	5.69	5.17	4.77	3.87	3.27	(44)	
	RadStd	0.26	0.29	0.46	0.32	0.37	0.50	0.35	0.35	0.35	0.29	0.21	0.27	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	+0.36	N/A	N/A	+0.36	N/A	N/A	N/A	N/A	+137	+102
Regional (32 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-4	-46	+173	+120

Nomenclature: See separate page